

Main Contributions

The Cycle-Constrained Aerial Suppression Base and Water Point Location Problem:
Joint Siting under Uncertainty with an Application to Cundinamarca, Colombia

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1. We introduce the cycle-constrained aerial suppression base and water point location problem, to our knowledge the first model that sites aerial bases and water refill points jointly, with aircraft productivity endogenous to the base to fire to water refill cycle rather than to base-to-fire distance alone.
2. Containment is modeled physically, liters delivered against a fire growth requirement, so the two-stage stochastic program with a mean-risk conditional value-at-risk (CVaR) objective needs no historical suppression outcome data, which do not exist for Colombia, and its only nonlinearity admits an exact McCormick linearization.
3. A fix-and-optimize matheuristic whose reduced subproblems are exact by construction solves the full 5,449-point water catalog, a scale at which the direct model cannot even be built, with a random destroy operator that provably escapes the local optima of purely geographic neighborhoods.
4. The case study is built entirely from open, verified sources around Colombia's real Firehawk acquisition, from satellite fire detections to sample average approximation scenarios with a statistical optimality gap of 3.4 percent, and is fully replicable from the companion repository.
5. The computational study yields policy-relevant structure: deployment is fleet-limited before it is physics-limited, sequential bases-first planning matches the integrated optimum in every certifiable regime, and the stable half of the optimal plan is a short, nearly free list of named water points, actionable before any aircraft arrives.